Mohammad Jahidul Islam Tushar

Deputy GIS Specialist

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Professional Summary

GIS and Remote Sensing specialist with strong expertise in spatial data quality management, geospatial database validation, and large-scale mapping project coordination. Experienced in leading multidisciplinary teams and ensuring spatial accuracy, topology compliance, and structured QA/QC frameworks for complex geospatial datasets. Demonstrated ability to improve workflow efficiency through structured automation approaches and systematic validation processes. Proven contributor to government, research, and private-sector geospatial initiatives requiring precision, reliability, and scalable data management solutions.

Core Competencies

GIS & Remote Sensing	ArcGIS, ArcGIS Pro, QGIS, ERDAS Imagine, ENVI, IDRISI Selva, Google Earth Pro
UAV & Photogrammetry	Pix4D, Agisoft Metashape, Orthophoto generation, Point cloud processing, DEM/DSM creation
Spatial Analysis & Modeling	Cartographic design, Geospatial modeling, LULC classification, Change detection, Spatial & non-spatial analysis
Automation & Programming	Python, Script Adaptation & Automation (AI-assisted), Batch Processing
Data Management	Spatial databases, Data quality control (QA/QC), Data analytics, Multi-source data integration
Visualization & Reporting	Tableau, SPSS, Adobe Illustrator, Adobe Photoshop, MS Office

Professional Experience

Deputy GIS Specialist | Sheltech Consultants (Pvt.) | Dec 2024 – Present | Dhaka, Bangladesh | Full-time

- Managed digitization QA/QC for mouza mapping and spatial databases, maintaining 90–95% data integrity, positional accuracy, and project compliance.
- Performed QC verification on geo-referencing, edge-matching, topology validation, and attribute accuracy for 2000+ map sheets using satellite and drone imagery, achieving <5% error rates and reducing rework by 35%.
- Coordinated quality assurance and updates for land-use/land-cover databases by validating and integrating multi-source satellite datasets for accurate land zoning outcomes.
- Developed Python scripts for automated QC validation, error detection, and batch processing (AI-assisted), improving QA efficiency by 30–40% and shortening database finalization timelines.

GIS Data Quality Manager | Institute of Water Modelling (IWM) | Feb 2023 – Nov 2024 | Dhaka, Bangladesh | Full-time

- Led teams of 15+ including the cartographic team, printing, digitization units, ensuring spatial accuracy and consistency across Dhaka and Barishal Division with <7% error rate.
- Conducted rigorous QA/QC and validation of digitized maps and GIS databases, achieving 85-90% overall data accuracy and preventing costly rework in project.
- Implemented geospatial modeling and automation scripting solutions, streamlining workflows and boosting team operational efficiency by ∼25%.

GIS Expert | Group Mappers (DGHS: National Rabies Control Program) | Feb 2019 – Jun 2021 | Dhaka, Bangladesh | Full-time

- Delivered nationwide GIS/RS training to 150+ participants, improving field data accuracy by 30% and strengthening rabies surveillance.
- Managed Kobo Collect mobile data systems and GIS servers, enabling real-time tracking of 700+ field reports; supported field teams, and ensured data synchronization, quality control, and server management.
- Generated disease-risk and health facility maps to support policy decisions and accelerate outbreak response planning.

GIS Business Development Executive | Shahriar GIS School | May 2019 – Aug 2024 | Dhaka, Bangladesh | Part-time

- Managed online workshops, website updates, social media campaigns, and promotional design, increasing course enrollment by 50%.
- Coordinated ambassador programs and built strategic partnerships to expand training reach and visibility in the GIS education sector.
- Analyzed marketing and social media performance data to optimize strategies, boosting engagement and conversions by 40%.

Technical Consultant | Centre for Environmental and Geographic Information Services (CEGIS) | Mar 2021 – Apr 2021 | Dhaka, Bangladesh | Full-time

- Managed map procurement for the National Population Census 2022.
- Reviewed, processed, and converted maps into required digital formats.
- Performed georeferencing to ensure spatial accuracy of census maps.

Executive Management Team Member | Group Mappers | Aug 2020 – Oct 2020 | Dhaka, Bangladesh | Part-time

- Conducted data analysis, processing, and visualization for project deliverables.
- Developed implementation methodologies and coordinated with team members to meet project goals.

Research Assistant | Mohammad Abdul Quader Research Group | Jan 2019 – Feb 2021 | Dhaka, Bangladesh | Part-time

- Delivered GIS and remote sensing analysis for multiple research projects over two years.
- Strengthened research outcomes by applying improved methodologies and quality data processing.
- Analyzed and visualized spatial and statistical data using Excel, SPSS, ArcGIS, QGIS, and ERDAS Imagine.
- Supported technical reporting and documentation for project deliverables.

Education

M.Sc. Geography & Environment CGPA 3.57/4.00
Jagannath University, Dhaka, Bangladesh 2019 – 2020
Applied GIS, Remote Sensing, Digital Image Processing

B.Sc. Geography & Environment CGPA 3.36/4.00
Jagannath University, Dhaka, Bangladesh 2016 – 2019
GIS Fundamentals, Cartography, Geospatial Analysis

Academic Fieldwork & Survey Projects

As part of B.Sc. Geography & Environment studies at Jagannath University, conducted field surveys and data collection across Bangladesh under faculty supervision. Gained practical experience in GPS-based spatial data acquisition, socio-economic/environmental assessments, and on-site observation.

- **Field Study 2016** (May 26–28): Household survey, cultural phenomena, and physical feature identification in Lawachara, Sreemangal, Moulvibazar.
- **Field Study 2017** (May 1–5): Socio-economic dynamics, security challenges, and coping mechanisms of Rohingya population in Ukhia, Cox's Bazar.
- **Field Study 2018** (Nov 18–20): River habitat survey in the Karatoya River basin, Panchagarh, Nilphamari, and Lalmonirhat.
- **Field Study 2019** (Oct 9–13): Land-use and environmental survey in Cox's Bazar region, including vegetation change detection (Ukhia), water level/quality analysis (Moheshkhali), and climate change impacts on livelihoods (Chakaria Upazila).

Publications

- Tushar et al. (2022). Land Use-Land Cover and Change Detection post-classification of Dhaka City Corporation Area (1972–2020). *Jamal Nazrul Islam National Conference 2022*, Chittagong, Bangladesh.
- Quader et al. (2022). Factors of land cover change of Dhaka city through geospatial techniques. *Earth Systems and Environment*, 5(3), 105–119.
- Tushar et al. Disaster risk reduction through mangrove management: A case study of Sundarbans and surroundings. *Jagannath University Journal of Life and Earth Sciences*. (In progress)

Professional Activities & Contributions

- **GIS Trainer** Geoinformatics Earth Observation Research Center (`https://youtube.com/@georc`) • practical GIS and Remote Sensing video tutorials
- **Volunteering & Community Mapping**
  - **Group Mappers** — Participated in crowdsourced GIS mapping projects (COVID-19 tracking, Dengue risk mapping)
  - **Green Voice** — Supported river conservation activities through geospatial data and mapping

Trainings & Certifications

- Cadastral Map Digitizing Using ArcGIS – GEOMARK LIMITED (Sep 2022)
- Introduction to NASA's "Black Marble" Night Lights Data – NASA (Dec 2020)
- Web GIS and Web Mapping: Basics of Open Source Platforms – Udemy (Nov 2019)
- The Crossroad of Exploring Geospatial Science – MORU (Aug 2022)
- GIS for Beginners: Introduction to GIS – GroupMappers (May 2019)
- GIS Software Technology Conference 2020 – SuperMap GIS (Oct 2020)

References

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